NI BiomedicalToolkit을 이용한 DICOM 이미지 처리 및 3D 시각화

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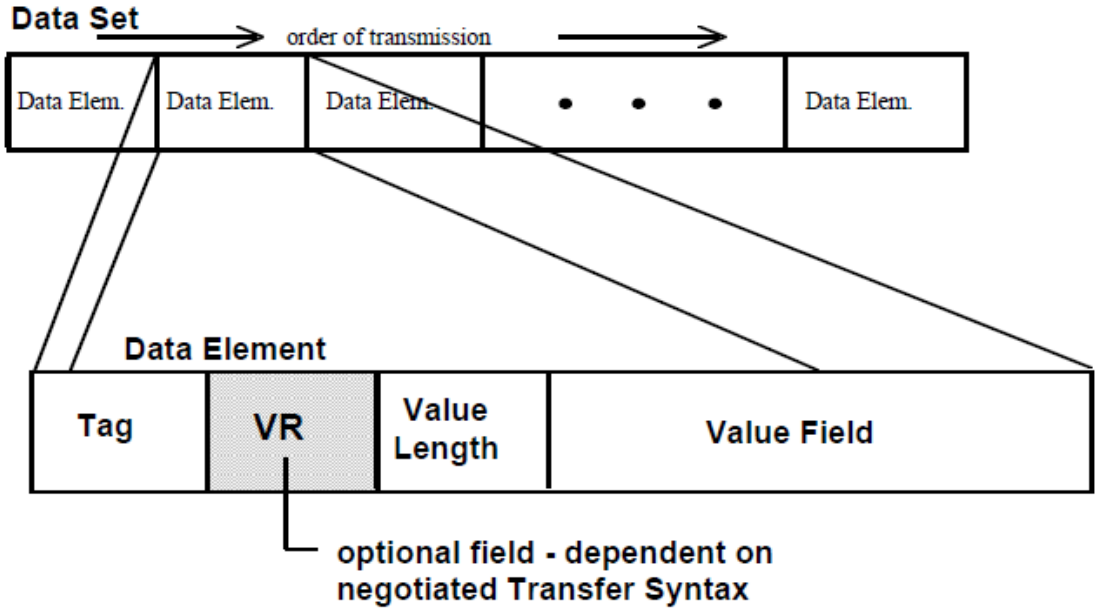
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DICOM 정의

- Digital Image and Communication on Medicine의 약자
- 1985년 국제 방사선학회가 정한 표준 프로토콜
- 의료 영상을 교환하고 구성하는 방법과 그에 관련된 정보들을 기술한 자세한 명세



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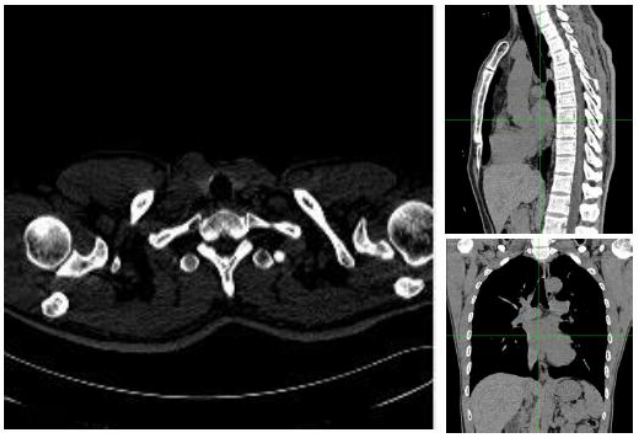
DICOM 특징

- CT,MR, US,CR,NM을 비롯한 각종 디지털 영상 장비가 만들어 낸 영상 정보 전송
- 환자의 검사 정보들에 대한 질의, 검색
- 네트워크에 연결된 영상 출력 장비와의 정보 교환
- CD와 같은 디지털 기록 매체를 이용한 영상 정보 교환
- 업체와 업체 간의 모든 장비에 대해 호환이 가능한 영상을 얻기 위하여 대부분 의료기 업체들이 DICOM 표준을 수용하고 있음.

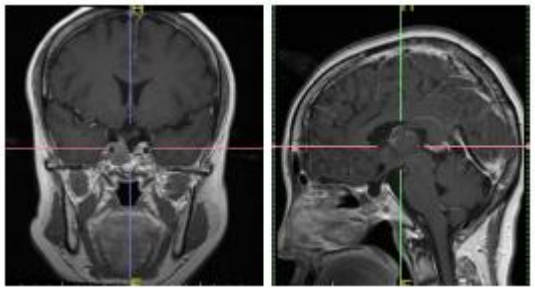
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의료 영상 종류

- CT(Computerized Tomography)
- MRI(Magnetic Resonance Image)
- PET(Positron Emission Tomography)
- SPECT(Single-Photon Emission Computed Tomography)
- US(Ultrasound Sonography)

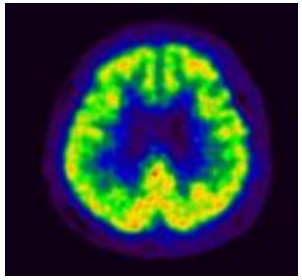


CT

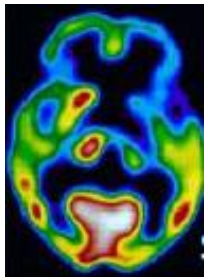


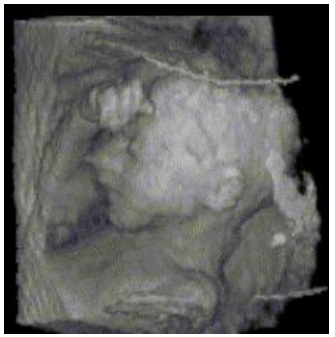
MRI

PET



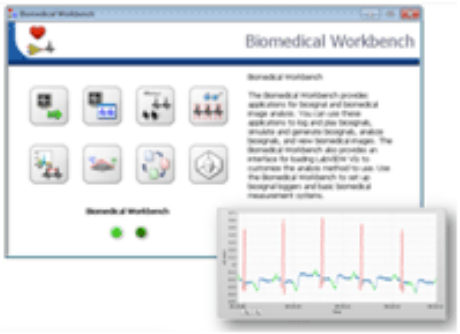
SPECT





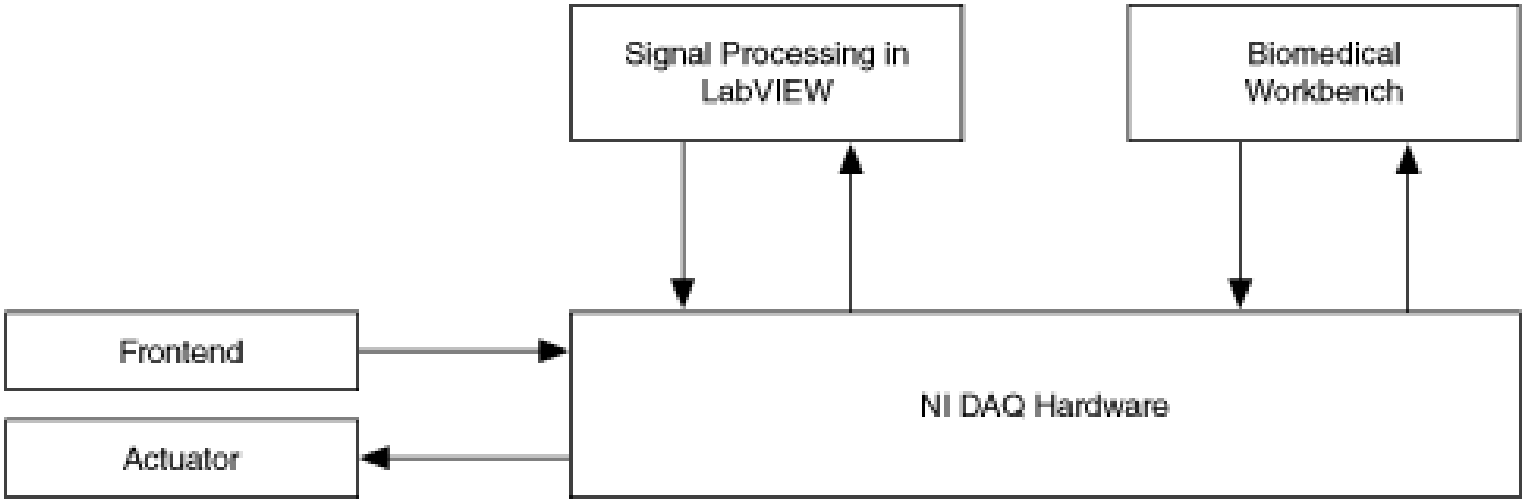
US

NI Biomedical Toolkit이란



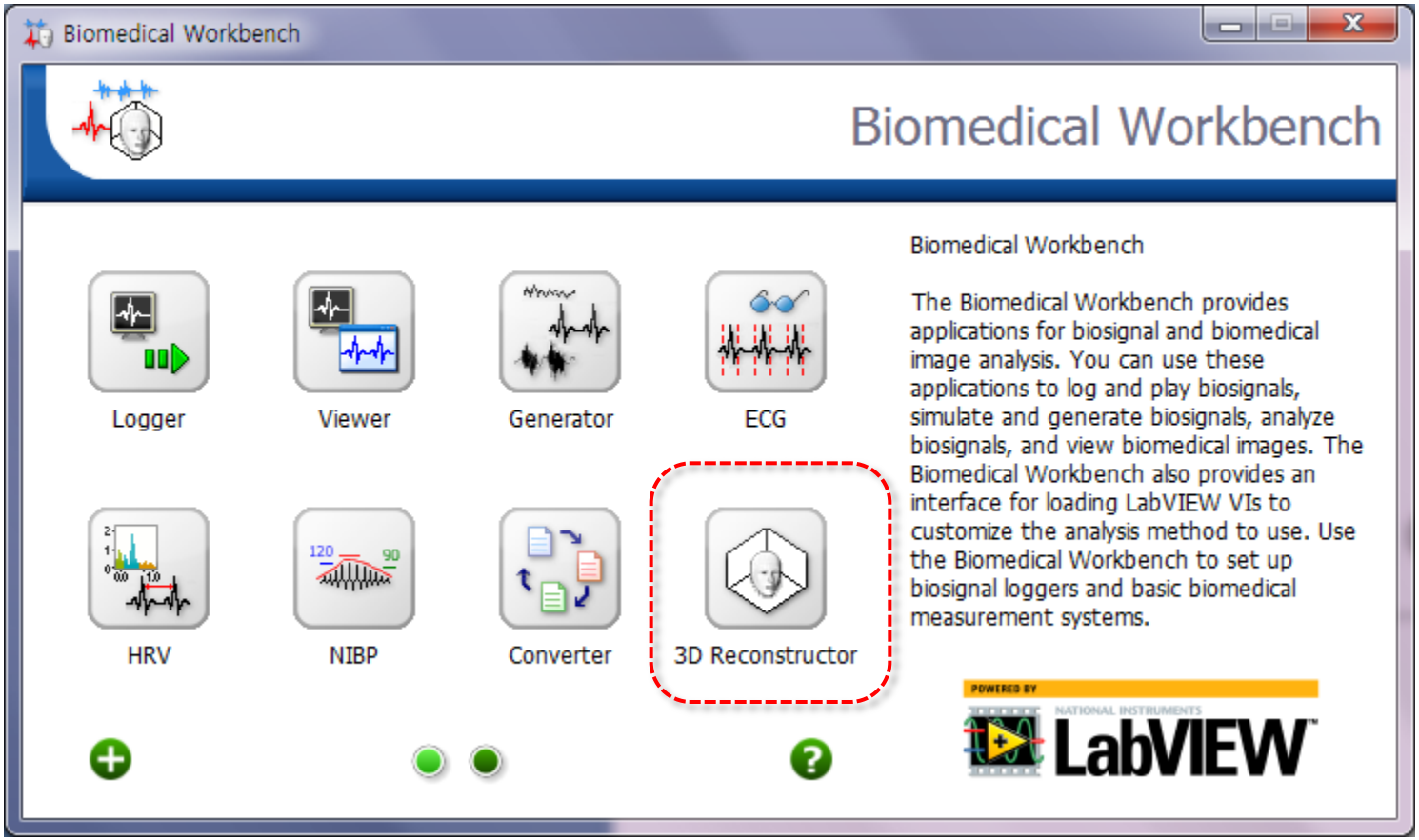
- 생체 신호를 디스크로 스트리밍하여 재생하고 분석할 수 있도록 전 기능을 갖춘 다채널 데이터 로거
- 일반 생리학 파일 포맷의 반입/반출을 간소화하는 파일 변환 유틸리티
- ECG 기능 추출, RR 주기 분석, 3D 이미지 재구성 등의 예제
- EEG, EMG, ECG 등을 위한 알고리즘이 있는 VI 라이브러리 포함
- 직접 설정 가능한 시작 관리자를 통해 사용자가 생성한 어플리케이션과 이이콘 추가
- NI ELVIS와 같은 NI 교육 플랫폼 및 대부분의 NI DAQ 제품과 긴밀히 작동

Biomedical analysis process

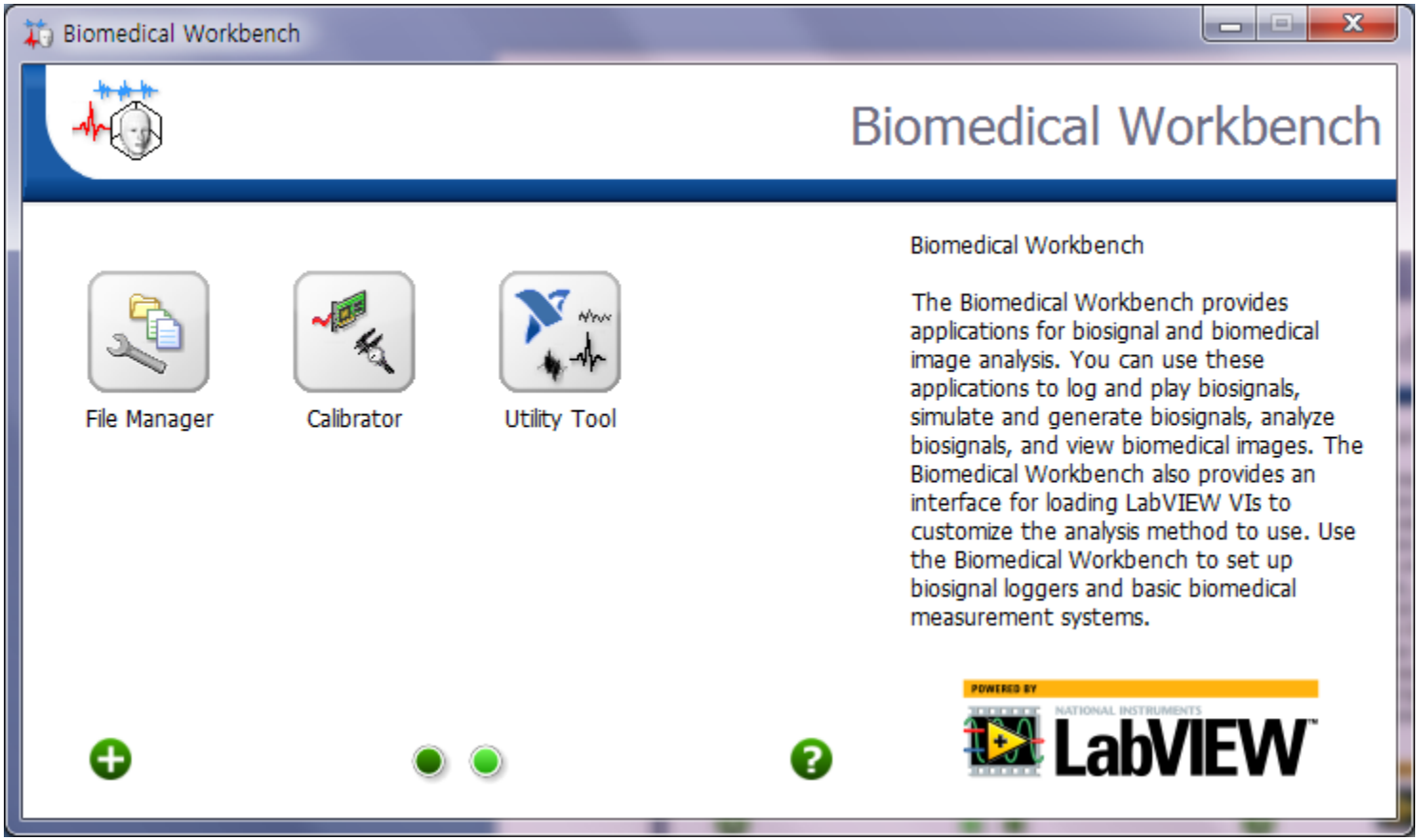


- Front End : 테스트 주체에 연결하기 위한 디바이스
- DAQ Hardware : Data를 취득하기 위한 하드웨어
- Signal Processing : biosignal과 biomedical 이미지 프로세싱
- Biomedical Workbench : 취득/프로세싱/추출/분석
- Actuator

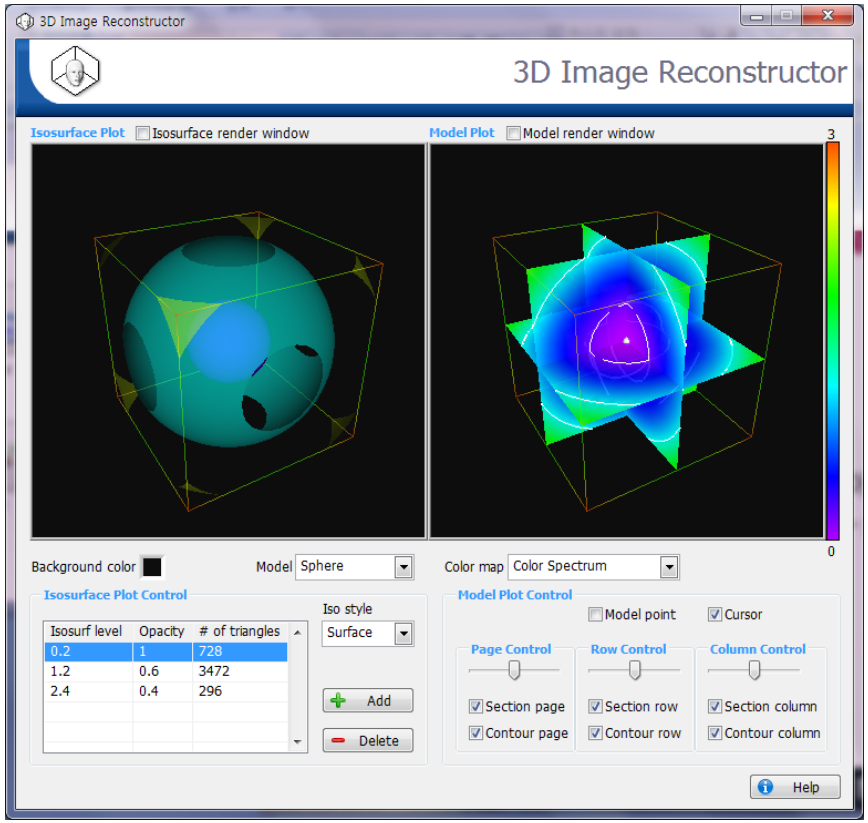
Biomedical Workbench

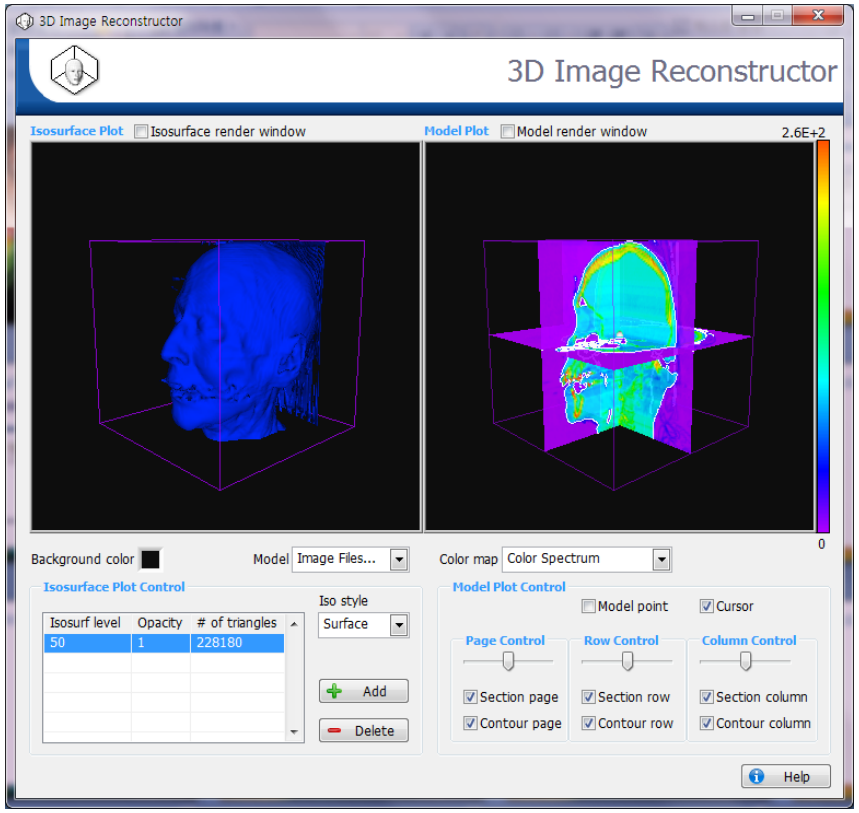


Biomedical Workbench

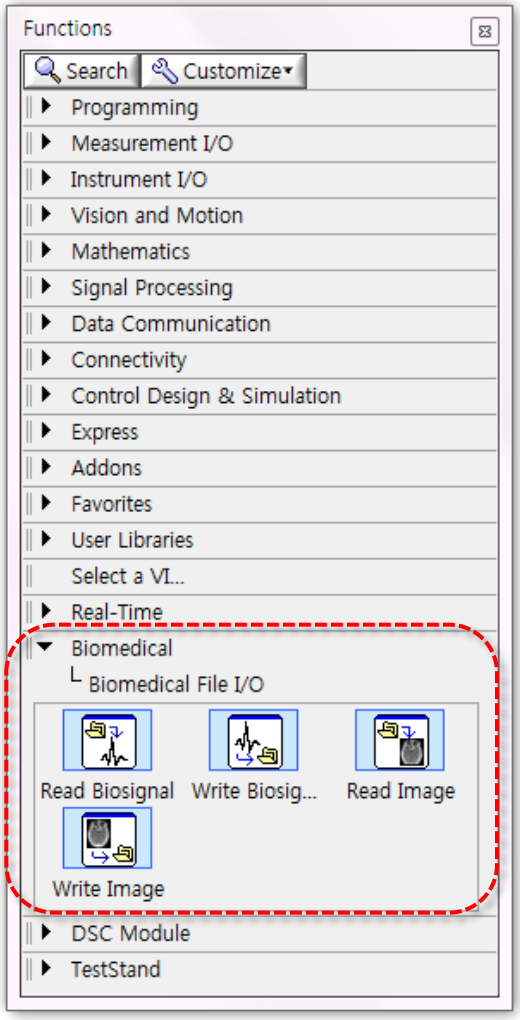


3D Image Reconstructor

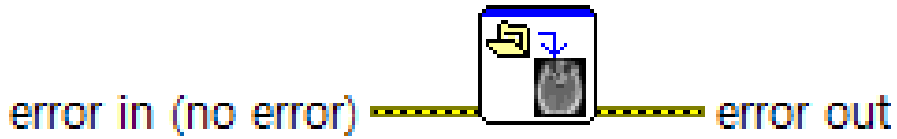




Biomedical File I/O



Read Image



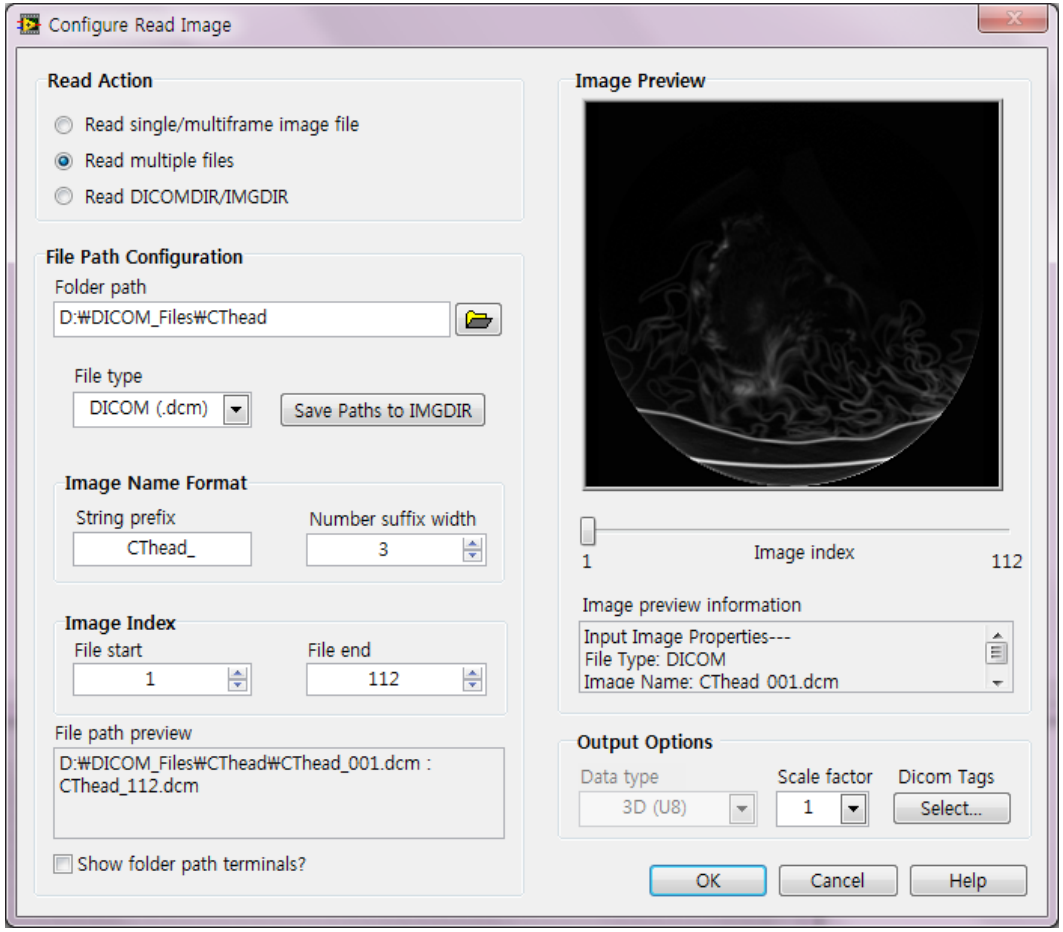
Reads images from different file types, such as BMP, JPEG, PNG, and DICOM.

Write Image

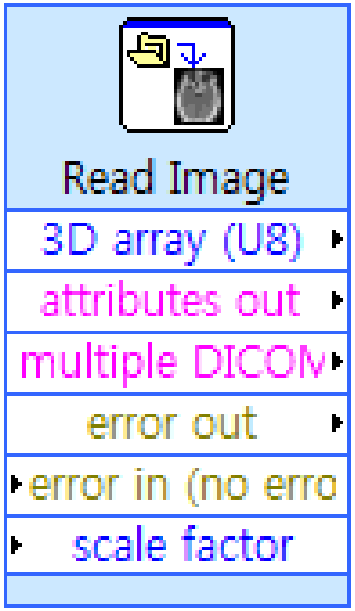


Writes image data to different file types, such as BMP, JPEG, PNG, and DICOM.

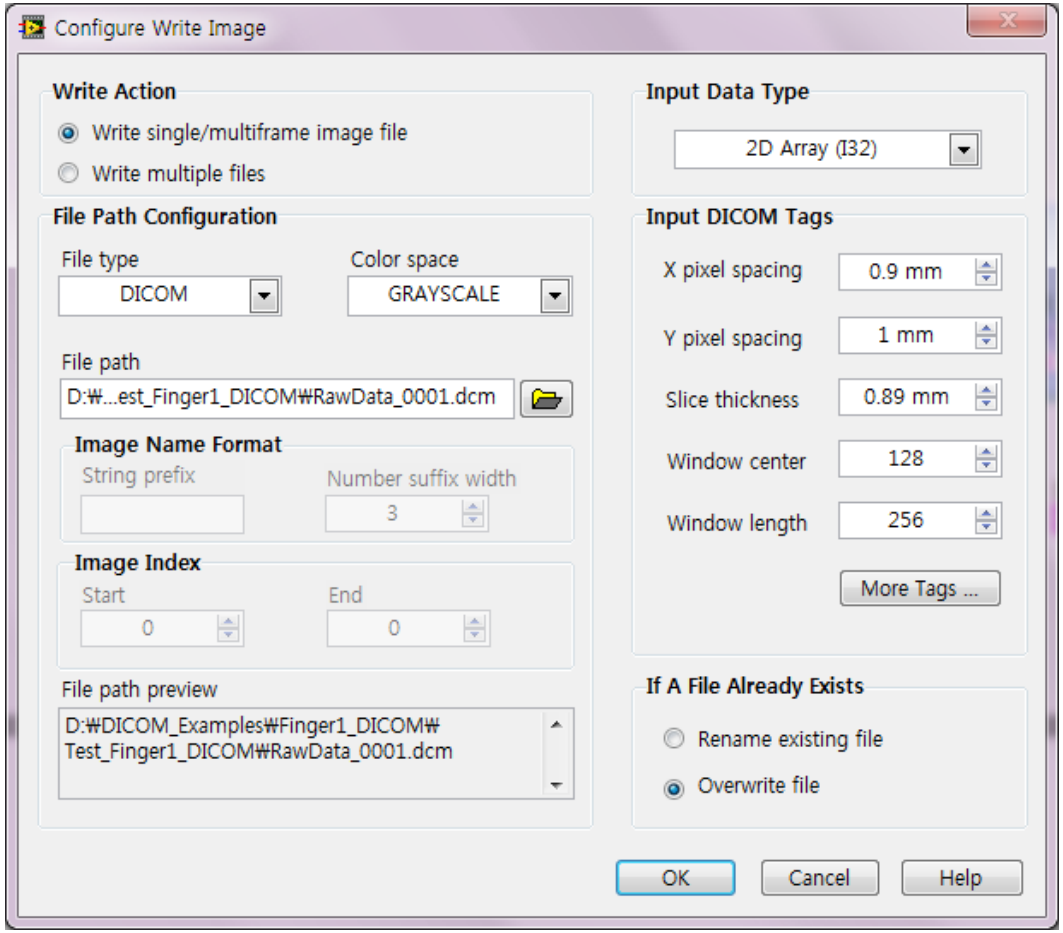
Biomedical File I/O



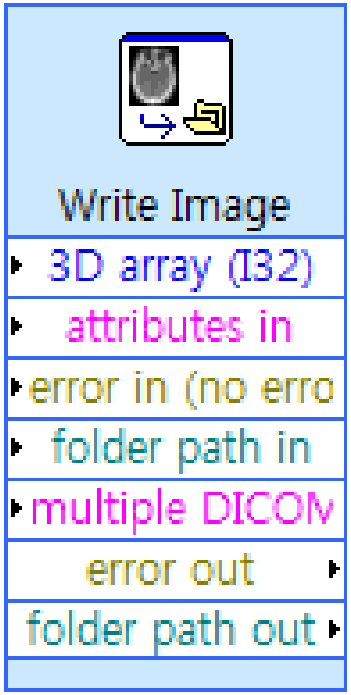
Read Image



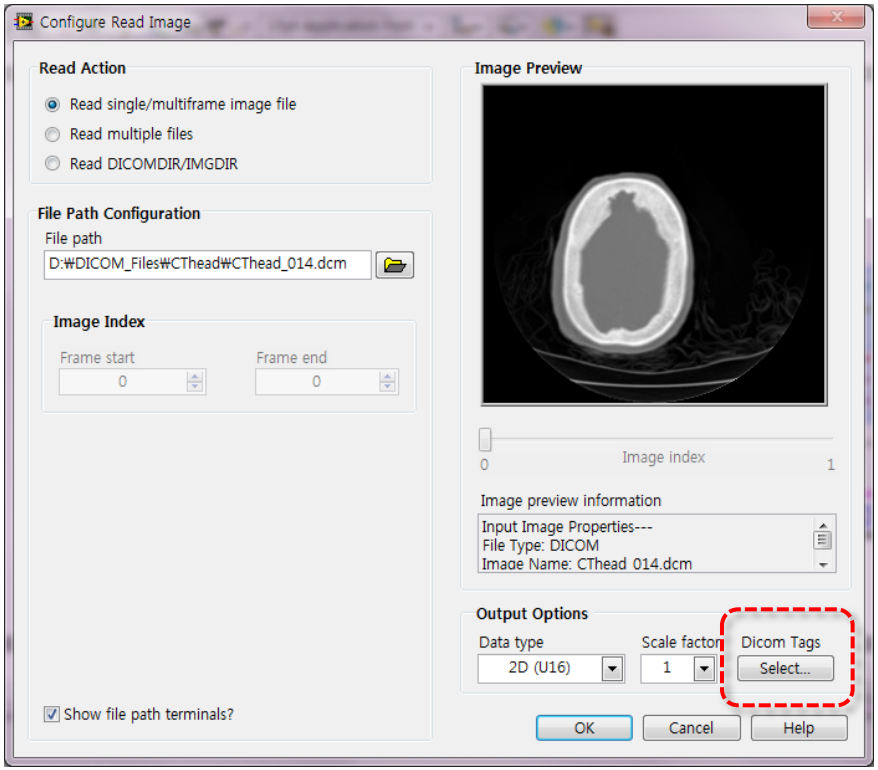
Biomedical File I/O

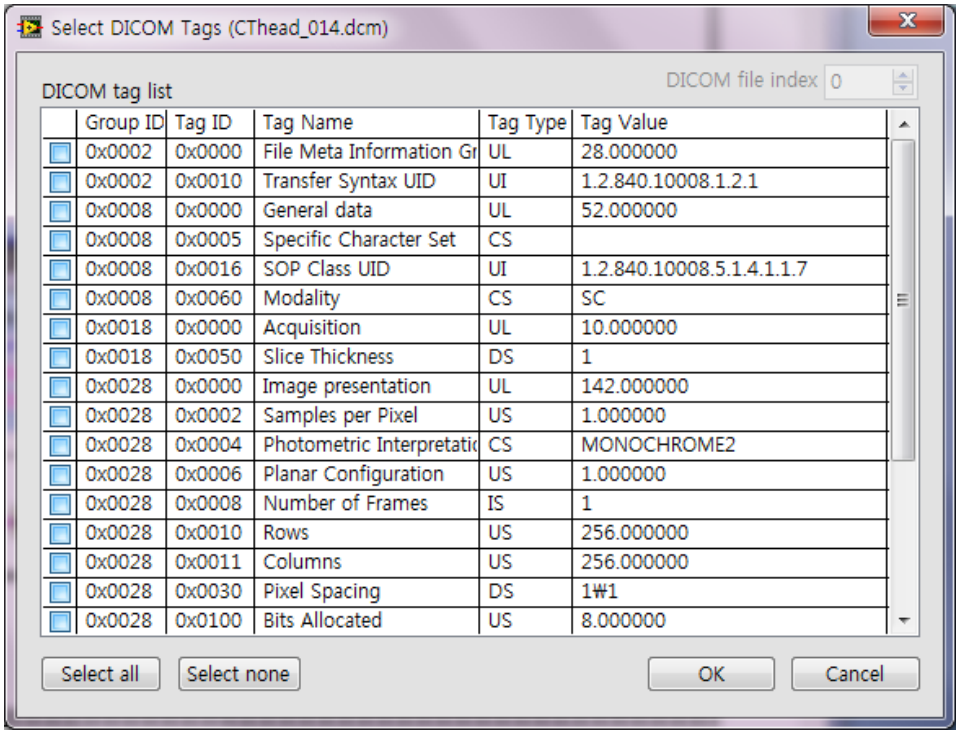


Write Image



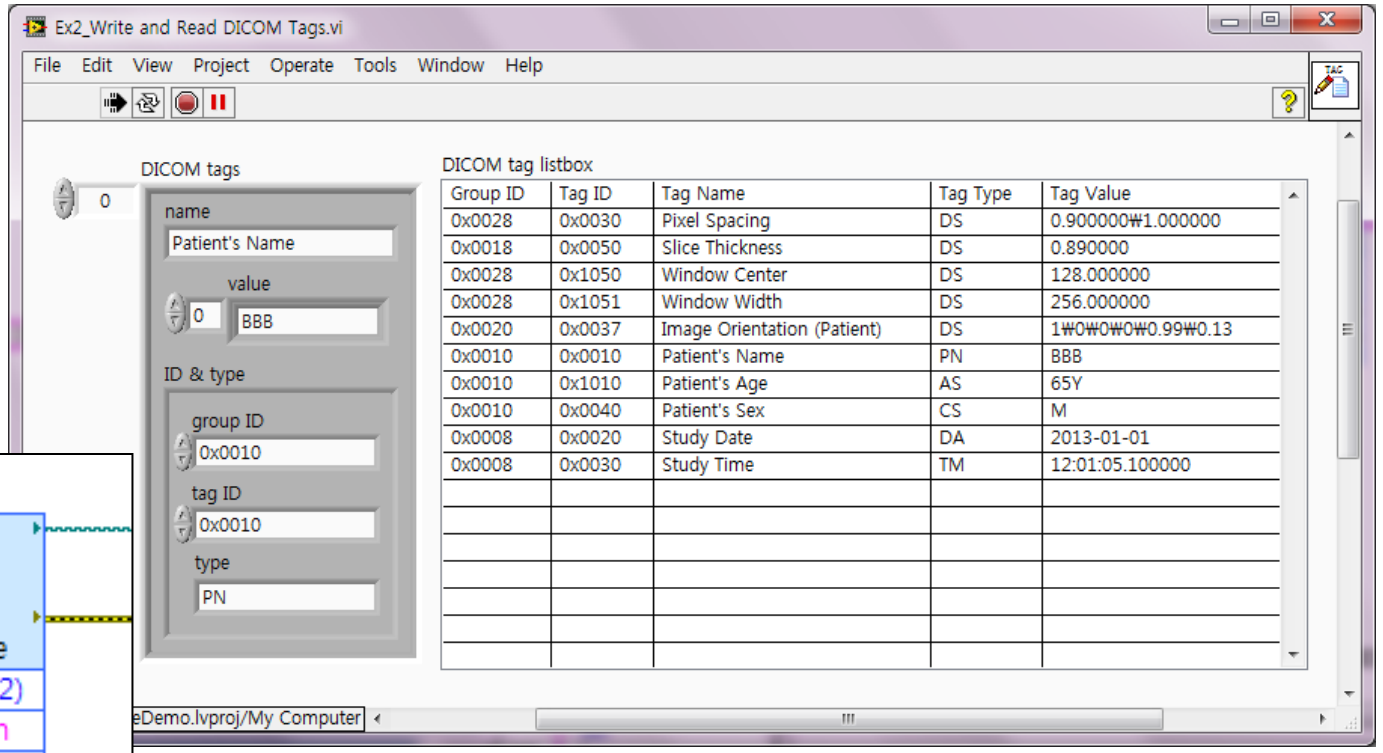
Read DICOM Tags

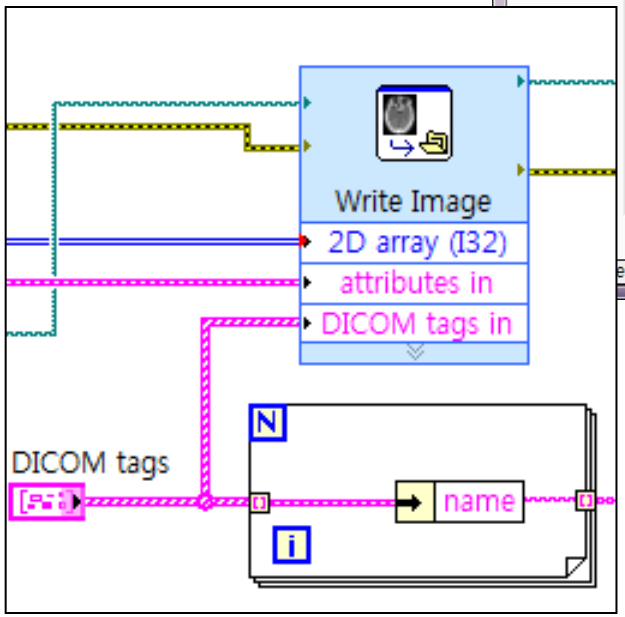




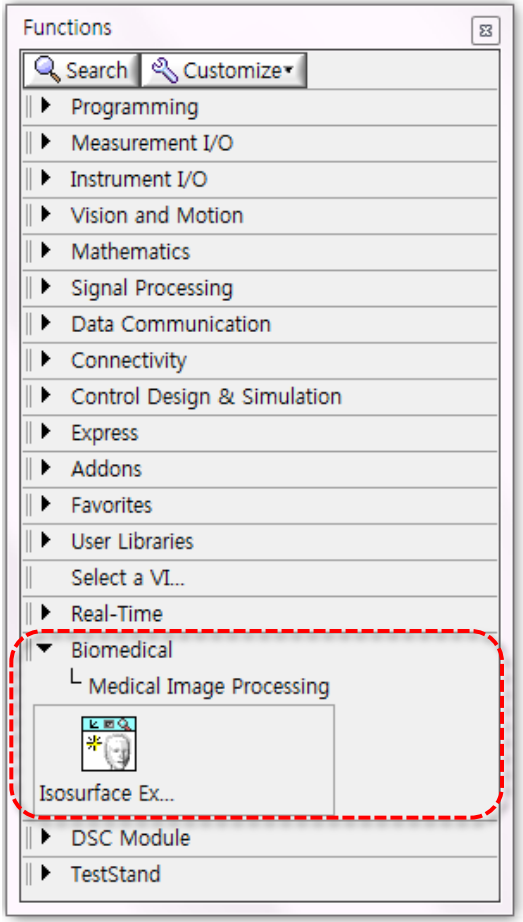
Dicom Tags

Write DICOM Tags

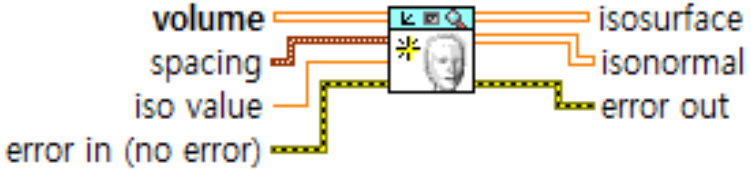




Medical Image Processing

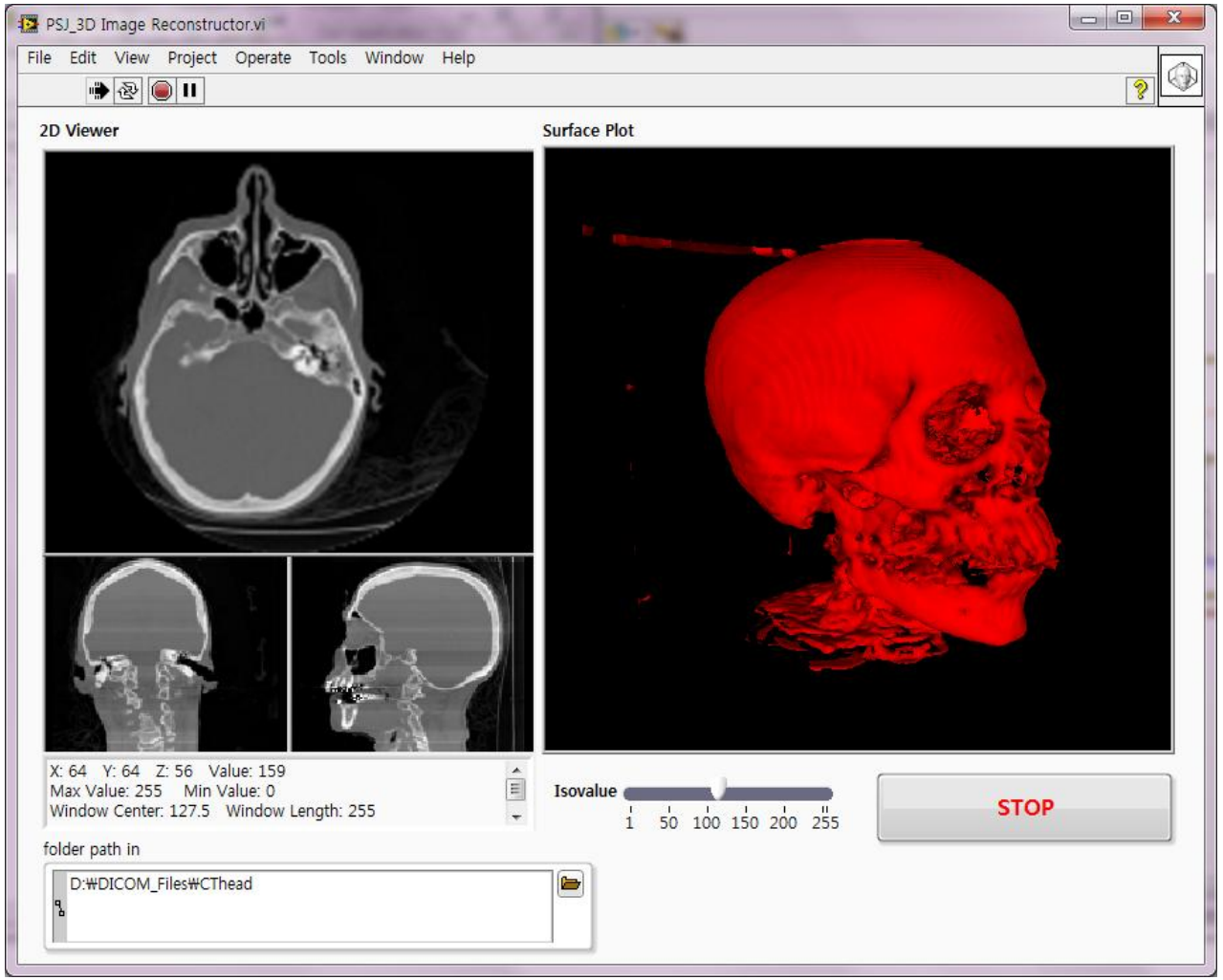


NI_BMTK_MedicalImage.lvlib:Medical Image Isosurface Extractor.vi

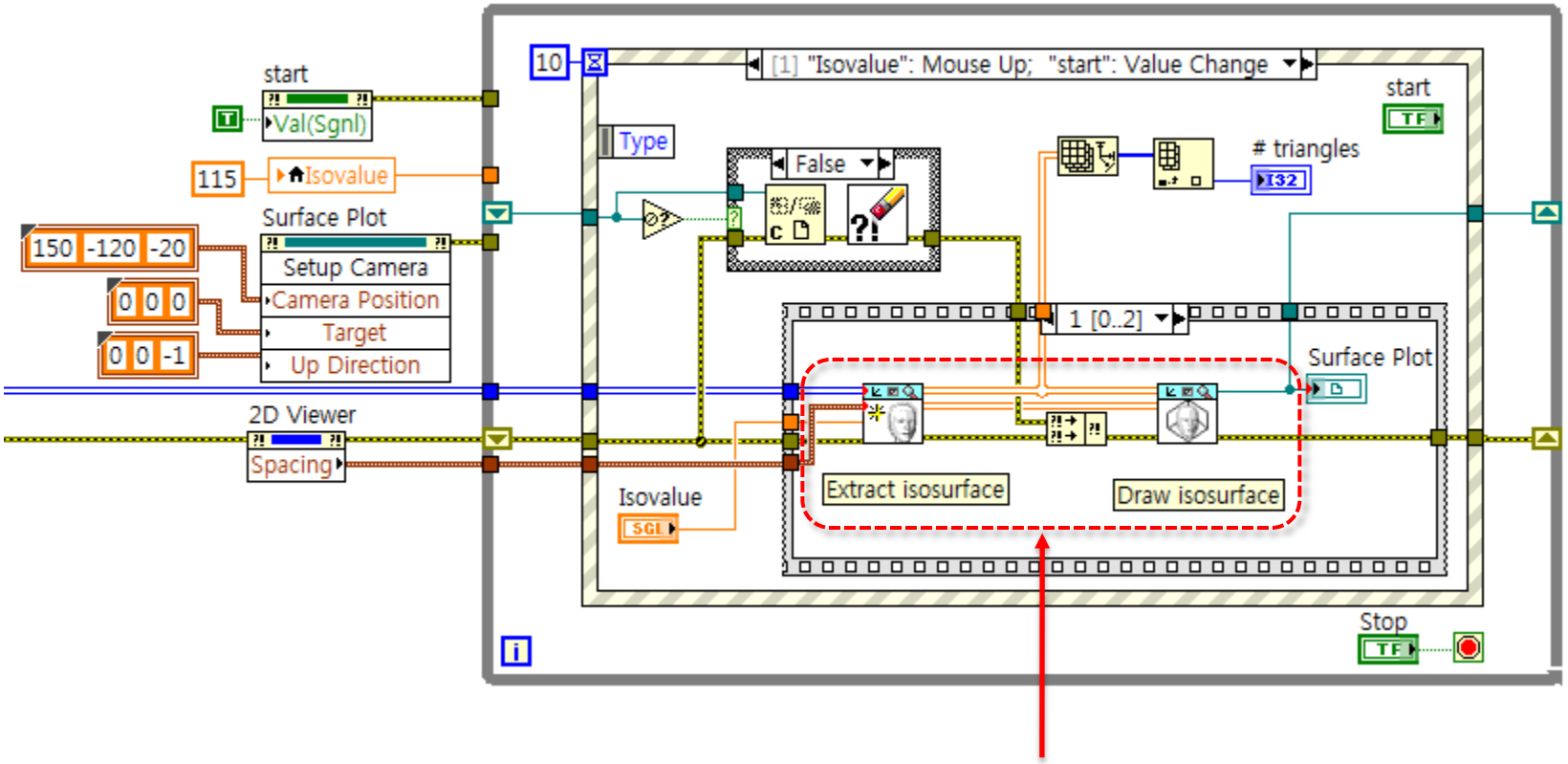


Extracts the isosurface and isonormal from a 3D volume data set.

Medical Image Processing (1)



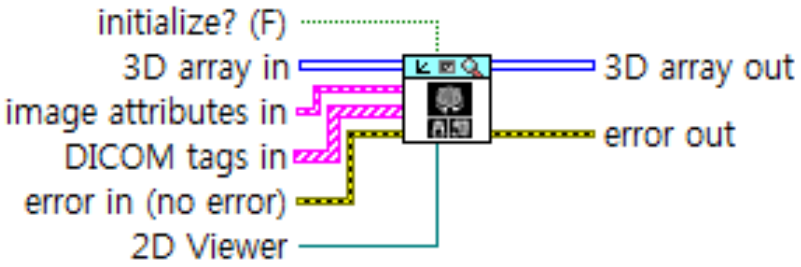
Medical Image Processing (1)



Extract & Draw ISO surface

Medical Image Processing (2)

NI_BMTK_Controls.lvlib:Medical Image 2D Viewer Helper.vi

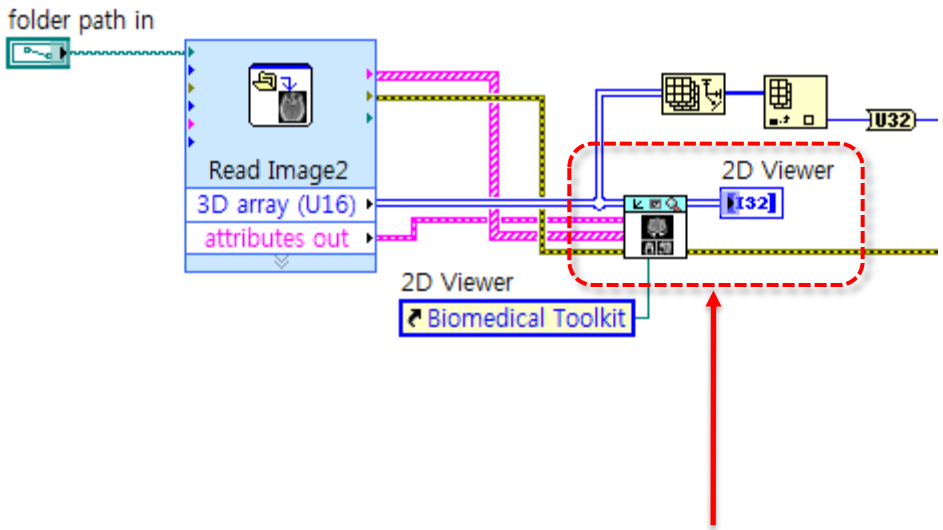


Detects the orientation of the input image from **DICOM tags in** and rotates the input image data according to the image orientation property. Also extracts common image attributes, such as color space, color map, spacing, and window size, from **image attributes in** and **DICOM tags in** and passes these attributes to the **2D Viewer** through the property nodes.

You can use the **Read Image** Express VI to read an image or image series and wire the data to this helper VI.

Wire data to the **2D array in** input to determine the polymorphic instance to use or manually select the instance.

Medical Image Processing (2)



Animate 2D slice image

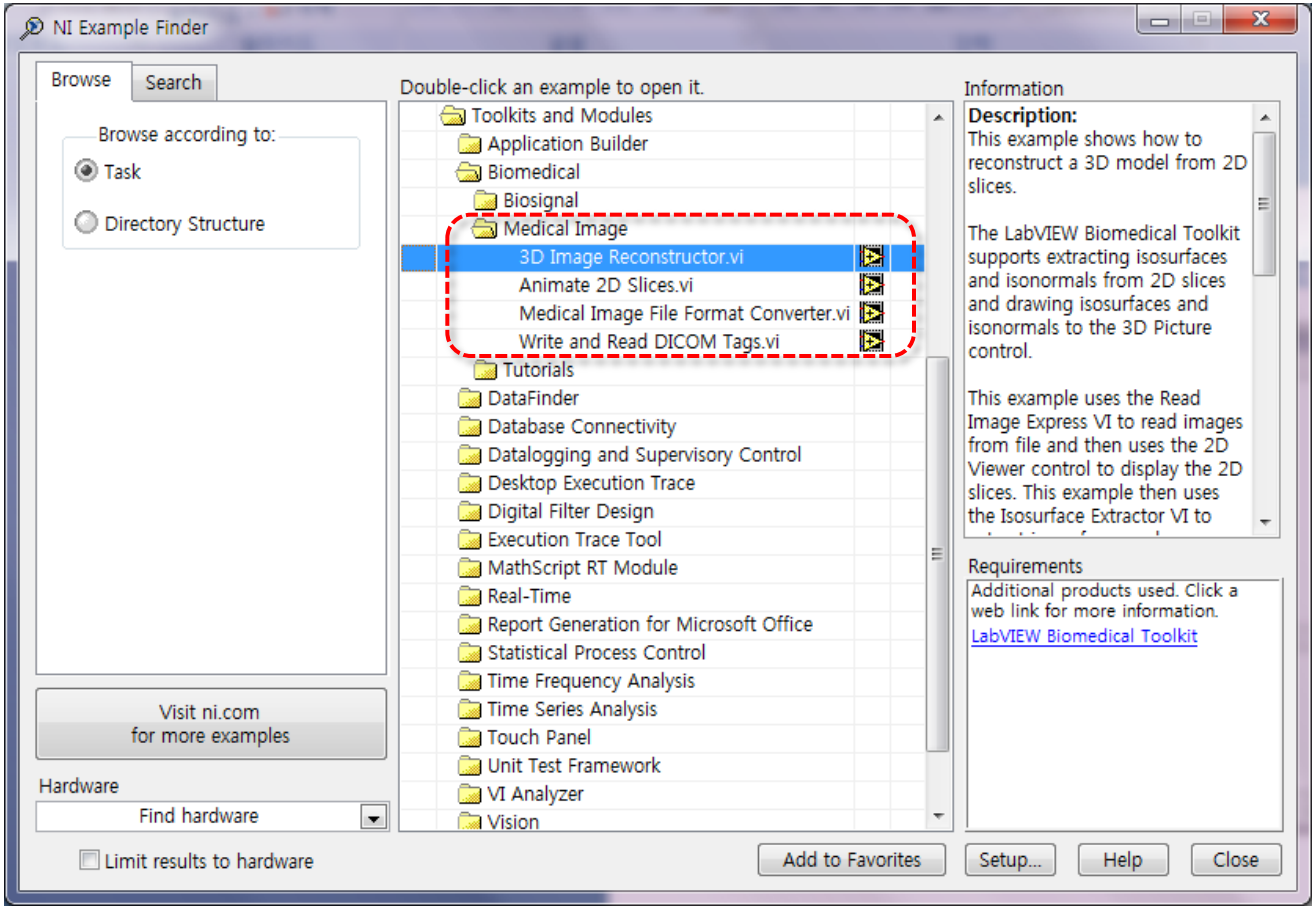


NI Biomedical Toolkit lvlib files

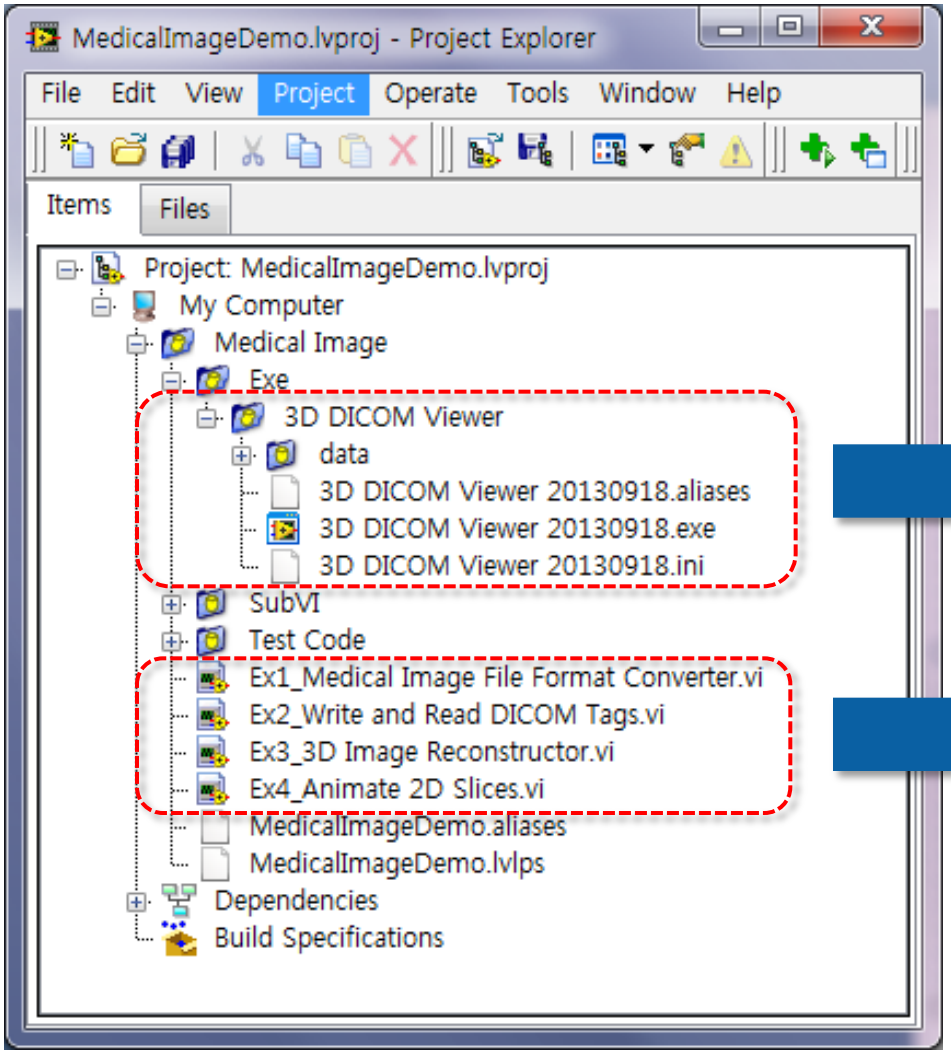
- C:\Program Files\National Instruments\LabVIEW 2013\vi.lib\addons\Biomedical
- NI_BMTK_Controls.lvlib : Biosignal / Medical Image
- NI_BMTK_Acquisition.lvlib : Express
- NI_BMTK_common.lvlib : Device IO / Image
- NI_BMTK_FileIO.lvlib : Express
- NI_BMTK_MedicalImage.lvlib : Medical Image
- NI_BMTK_Measurements.lvlib :

  Biosignal Preprocessing / ECG / EEG / EMG / NIBP

NI Biomedical Toolkit Example Codes



Demo



Development Program

Modify Example Codes

Q & A

